



Chapter 3

Internet of Things (IoT) & Robotics

Smart Systems, Embedded Devices, and Autonomous Computing

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Objectives

- Explain what IoT is and its key features
- Describe IoT architecture and different types of sensors
- Understand embedded systems and edge computing
- Explain robotics basics like kinematics and motion planning
- Describe how IoT integrates with AI and cloud computing



What is IoT?

- **IoT** stands for Internet of Things.
- It is a network of physical objects (things) that are connected to the internet. These objects have sensors, software, and other technologies that allow them to collect and exchange data without human involvement.
- Example: A smart light bulb you can control from your phone.



What is IoT?

IoT = Services + Data + Networks + Sensors

- **Sensors** collect information from the environment
- **Networks** send that information to other devices
- **Data** is the information collected and analyzed
- **Services** are the useful functions provided to users

Example: A fitness tracker collects your heart rate (sensor + data), sends it to your phone (network), and shows your health report (service).



Key Features of IoT

1. Connectivity – Devices can connect to the internet and each other
2. Data Collection – Sensors gather real-world information
3. Remote Monitoring – You can check devices from anywhere
4. Automation – Devices can act without human commands
5. Efficiency – Saves time, energy, and money



IoT Architecture

- IoT devices are built using 4 layers stacked on top of each other:

1. Sensing Layer: Collects data from environment
2. Network Layer: Transmits data between devices
3. Data Processing Layer: Analyzes data and makes decisions
4. Application Layer: Shows results and lets users interact



Sensing Layer

- This is the bottom layer that interacts directly with the physical world.
- Components:
 - Sensors – Detect changes in environment (temperature, motion, light, humidity, location)
 - Actuators – Perform actions (open a door, turn on a motor, switch off a light)
 - Sensor Hub – A central point that collects data from multiple sensors
- Example: In a smart home, a motion sensor detects someone walking in a room.



Types of Sensors in IoT

Sensor Type	What it detects	Example use
Motion sensor	Movement	Security alarm
Temperature sensor	Heat level	Smart thermostat
Humidity sensor	Moisture in air	Weather station
Light sensor	Brightness	Auto-dimming lights
Position sensor	Location	GPS tracker
Gas sensor	Smoke or gas	Fire alarm



Network Layer

- Acts as a communication channel to transfer data collected by the sensing layer to other devices or the cloud.
- Technologies used:
 - Wi-Fi – High speed, short range (home networks)
 - Bluetooth – Very short range, low power (headphones, smartwatches)
 - Zigbee / Z-Wave – Low power, used in smart home devices
 - LoRa – Long range, low data (farm sensors)
 - Cellular (4G/5G) – Wide coverage, high data (cars, city-wide sensors)



Data Processing Layer

- Takes raw data from sensors and analyzes it to make useful decisions.
- What happens here:
 - Data is cleaned and organized
 - Rules and logic are applied (if temperature $> 30^{\circ}\text{C}$, turn on AC)
 - Decisions are made locally or sent to the cloud
 - Some devices save past results to learn and improve
- Example: A smart thermostat learns that you like 22°C at 8 PM and automatically sets the temperature.



Application Layer

- This is the top layer that users interact with. It presents the results of data processing in a useful way.
- Examples of applications:
 - Smart home app (control lights, locks, cameras)
 - Healthcare dashboard (view patient heart rate)
 - Traffic management system (see live congestion)
 - Farming app (check soil moisture levels)
- User-centric: This layer is designed for people to easily understand and control their IoT devices.



What are Embedded Systems?

- An embedded system is a small computer built inside a device to perform a specific task.
- It has a processor, memory, and software, but it is not a general-purpose computer like a PC.
- **Characteristics:**
 - Designed for one or a few tasks
 - Low power consumption
 - Often small and cheap
- **Examples:** Microcontroller in a smartwatch, processor inside a smart TV, chip in a washing machine.



What is Edge Computing?

- Edge computing means processing data near the device that collected it, instead of sending all data to a central cloud server.
- **Benefits:**
 - **Faster response** – No delay from sending data far away
 - **Saves bandwidth** – Less data sent over internet
 - **More privacy** – Sensitive data can stay on device
- **Example:** A security camera detects motion locally and only sends video to the cloud when something important happens.



Edge vs. Cloud Computing

Feature	Edge Computing	Cloud Computing
Location	Near the device	Far away (data center)
Speed	Very fast	Slower due to distance
Processing power	Limited	Very powerful
Best for	Real-time decisions	Big data analysis, storage
Example	Smart home hub	Google Photos storage

Many IoT systems use both: edge for quick actions, cloud for heavy analysis.



IoT Applications – Real World Examples

Domain	Example
Smart Home	Smart lights, locks, thermostats
Smart City	Traffic control, waste management
Healthcare	Heart monitors, pill dispensers
Agriculture	Soil sensors, automated irrigation
Transportation	Connected cars, fleet tracking
Retail	Inventory tracking, smart shelves



What is Robotics?

- Robotics is the branch of technology that deals with the design, construction, operation, and use of robots.
- What is a robot?
 - A programmable machine that can perform tasks automatically or with some guidance. Robots can sense their environment, process information, and take action.
- Examples: Robot vacuum, warehouse robot, surgical robot, self-driving car.



Robotics Fundamentals – Kinematics

- Kinematics is the study of motion without worrying about forces. In robotics, it helps us understand how robot arms and joints move.
- Two types of kinematics:
 1. **Forward Kinematics** – You know the joint angles. You calculate where the robot's hand (end effector) will be.
Example: If elbow is bent 90° and shoulder is raised 30° , where is the hand?



Robotics Fundamentals – Kinematics

2. Inverse Kinematics – You know where you want the robot's hand to be. You calculate what joint angles are needed.

- *Example:* You want the robot hand to reach a cup on a table. What angles should each joint have?
- **Why it matters:** Inverse kinematics is harder but more useful. It is used in robotic arms, walking robots, and animation.



Motion Planning (Robotics)

- Motion planning is finding a path for a robot to move from a starting point to a goal point while avoiding obstacles.
- Steps in motion planning:
 - Sense the environment (using cameras or sensors)
 - Build a map of obstacles
 - Calculate a safe path
 - Execute the movement

Example:

- A self-driving car calculates a route to avoid pedestrians and other cars.
- A warehouse robot finds the best path to move boxes without hitting shelves.



AI-Driven Robotics

- What AI adds to robotics: Artificial Intelligence gives robots the ability to learn, adapt, and make decisions based on data.
- Capabilities AI brings:
 - Computer vision – Recognize objects, faces, and obstacles
 - Learning – Improve performance over time
 - Decision making – Choose the best action in complex situations
 - Natural interaction – Understand voice commands
- Example: A robot vacuum learns the layout of your house and remembers where furniture is



Smart Automation

- Smart automation combines IoT sensors, AI, and robotics to perform complex tasks without human intervention.
- **How it works:**
 - IoT sensors collect real-time data
 - AI analyzes the data and decides what to do
 - Robots or actuators carry out the action
- **Example in a factory:** A sensor detects a defective product on the assembly line. AI confirms the defect. A robotic arm removes the bad product. All without any human stopping the line.



Integration of IoT with AI

- **Why integrate IoT with AI?** IoT devices collect huge amounts of data, but raw data is not useful by itself.
- AI finds patterns, makes predictions, and creates insights from IoT data.
- **Examples:**
 - Smart thermostat – Learns your schedule and adjusts temperature automatically
 - Predictive maintenance – AI analyzes machine vibrations to predict when a part will fail
 - Smart grid – AI predicts electricity demand and balances supply



Integration of IoT with Cloud

- **Why integrate IoT with cloud?** IoT devices often have limited storage and processing power.
- The cloud provides unlimited storage and powerful computing.
- **Benefits:**
 - Store data from millions of devices
 - Run complex analytics and machine learning
 - Access data from anywhere
 - Update device software remotely
- **Example:** A fitness tracker sends your daily steps to the cloud. The cloud stores months of data and shows you long-term trends on your phone.



Complete Integration – IoT + AI + Cloud

- All three technologies work together in a cycle.
 - **IoT devices** collect data from the physical world (sensors, cameras, etc.)
 - **Edge computing** handles urgent decisions locally
 - **Cloud** stores all data and runs powerful AI models
 - **AI** finds patterns, makes predictions, and sends decisions back
 - **IoT actuators** carry out those decisions
- **Real-world example – Smart traffic system:**
 - IoT cameras detect traffic (collect data)
 - Edge computer changes lights locally if emergency vehicle approaches (fast action)
 - Cloud analyzes traffic patterns over months (AI)
 - AI suggests better light timing to reduce jams



Advantages of IoT + Robotics

Advantage	One-Line Description
Efficiency	Tasks are completed faster and with less wasted time, energy, or materials.
Automation	Repetitive or boring tasks are performed automatically without needing human involvement.
Better decisions	Decisions are based on real data and analysis, not on guessing or intuition.
Safety	Robots and IoT systems can perform dangerous jobs so humans stay out of harm's way.
24/7 operation	These systems can run continuously, all day and night, without getting tired or needing breaks.
Cost savings	Reducing labor needs and human errors leads to significant long-term cost reductions.



Challenges of IoT + Robotics

Challenge	Description
Security	Hackers can break into IoT devices or robotic systems because many of these devices are connected to the internet and may have weak protection.
Privacy	IoT devices collect massive amounts of personal information, often without users fully knowing or agreeing. This data can be misused.
Complexity	Many devices and systems to manage
No standards	There is no single, universal standard that all IoT devices must follow. Devices from different brands may not work together
Cost	Initial setup can be expensive
Reliability	IoT and robotic systems depend on many components working perfectly together. If any single part fails (power, network, sensor, cloud server), the entire system may stop working or behave incorrectly.



Thanks